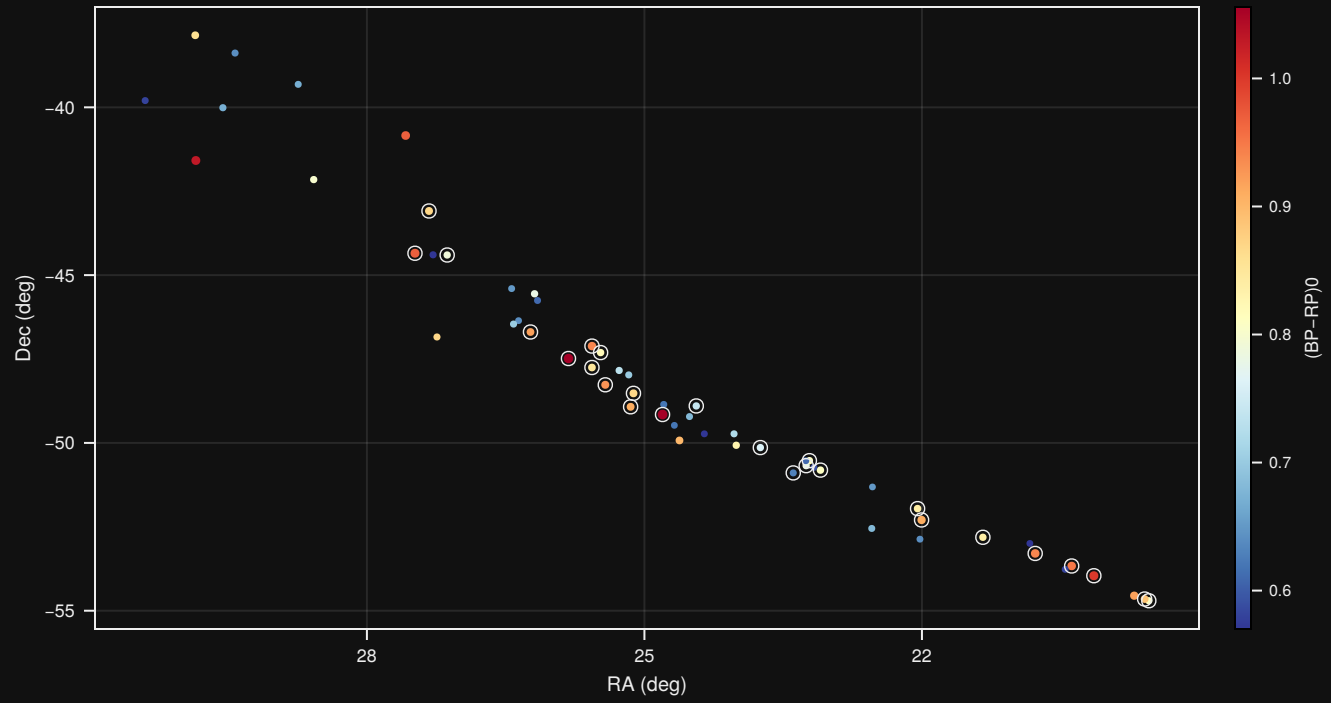
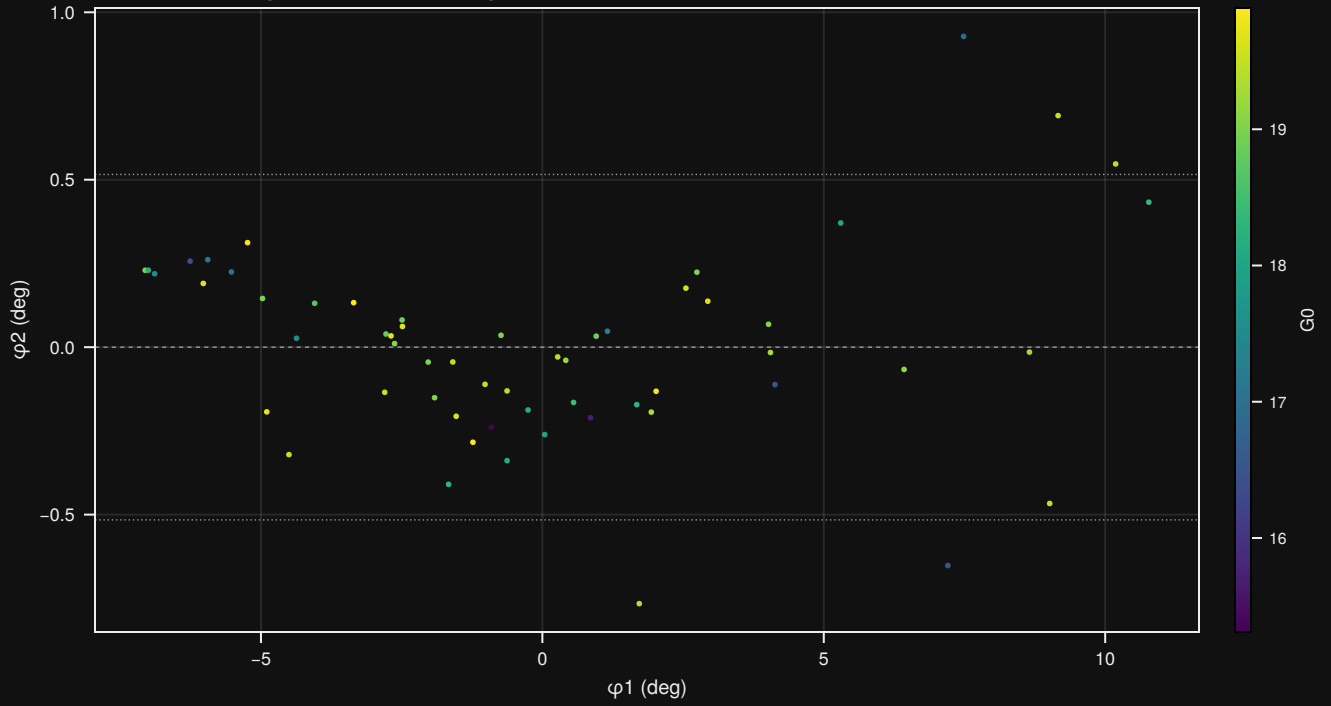
Phoenix

MEMBERSHIP	
Stars in catalog	58
with Gaia GSP-Phot	20 (34%)
with radial velocity	26
Stream id (sID)	5
GEOMETRY · great-circle fit	
Centroid RA, Dec	24.8, -48.9 deg
Centroid l, b	281.0, -66.3 deg
Pole RA, Dec	310.0, 13.0 deg
Length (φ1, 2–98%)	17.0 deg
Width σ(φ2), robust	0.26 deg
KINEMATICS	
pmRA* med ± σ	2.74 ± 0.14 mas/yr
pmDE med ± σ	-0.09 ± 0.21 mas/yr
μφ1* med ± σ	0.82 ± 0.29 mas/yr
μφ2 med ± σ	-2.58 ± 0.13 mas/yr
Parallax (cat.) med	0.067 ± 0.149 mas
Parallax (wtd, ZP-corr.)	0.072 ± 0.020 mas
Distance	d = 13.9 ± 3.9 kpc
Physical length	4.15 kpc
RV median ± σ(MAD)	51.6 ± 10.3 km/s
RV gradient dV/dφ1	-3.17 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	15.3–19.9 (19.0)
(BP−RP)0 median	0.80
GSP-Phot [M/H] (n=20)	-2.26 ± 0.82
GSP-Phot Teff median	5372 K
RUWE > 1.4	0.0 %
PM clusters (largest)	2 (24 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	58 | 58
n_RV catalog | paper	26 | 26
<[Fe/H]> paper	-2.70
Stellar mass (paper)	3000 Msun
Pole offset vs paper	0.8 deg

Sky map · catalog members (n = 58) · size ∝ brightness · ring = has RV

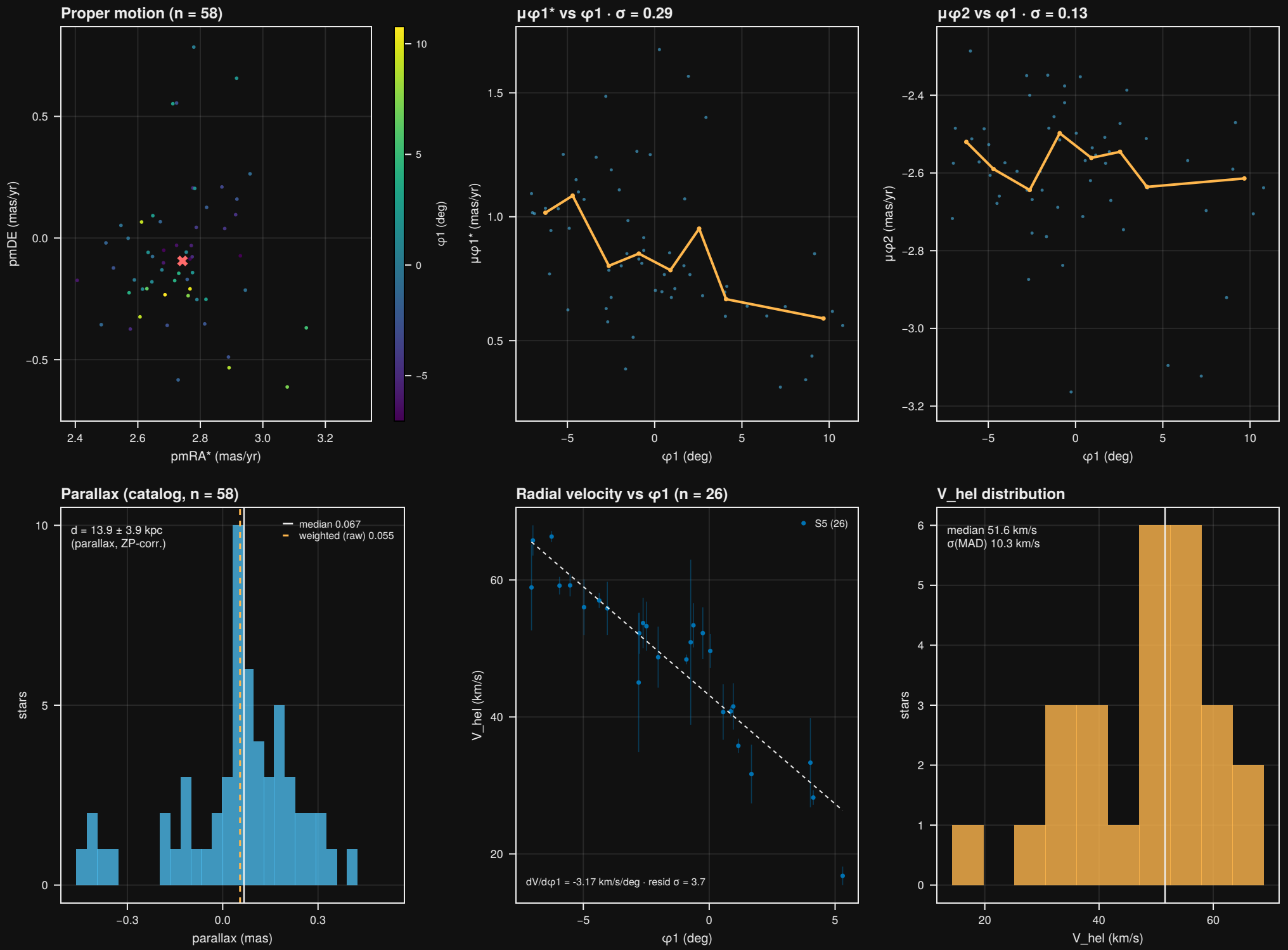


Stream frame (great-circle fit) · length 17.0° · width σ 0.26°



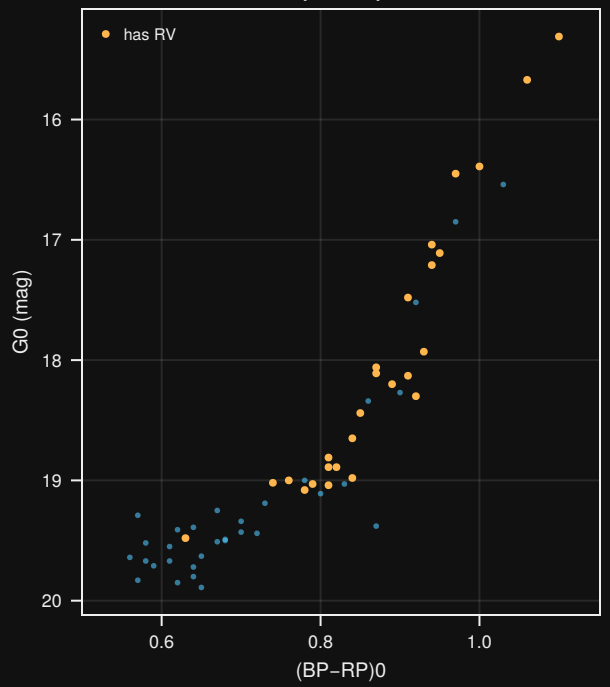
Phoenix

Kinematics · proper motion · parallax · radial velocity

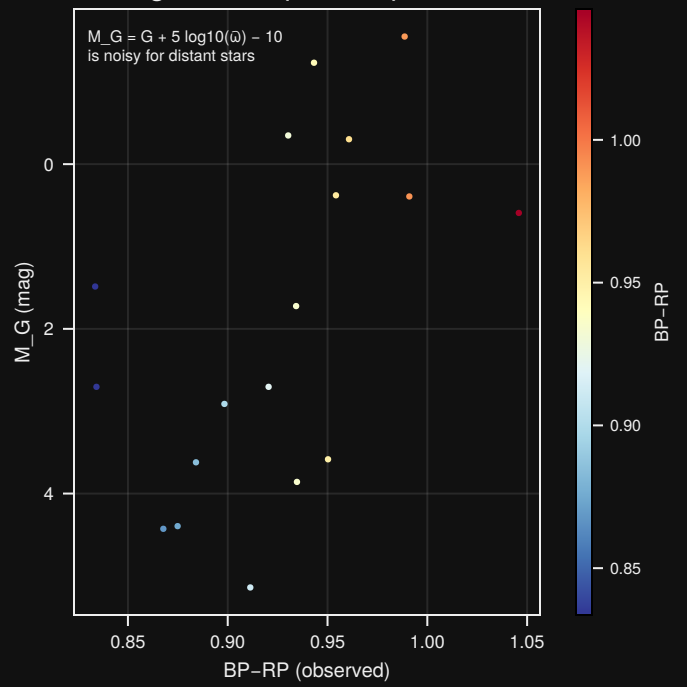


Phoenix

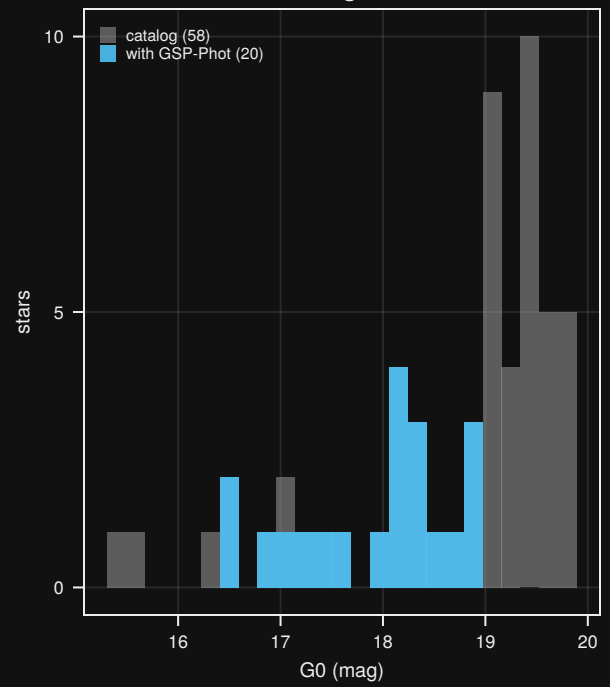
CMD · dereddened (n = 58)



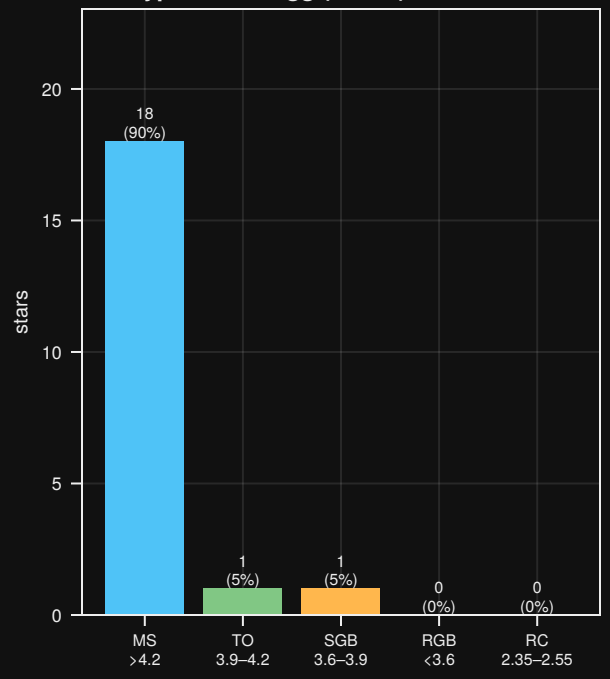
HR diagram · ϖ>0 (n = 18/20)



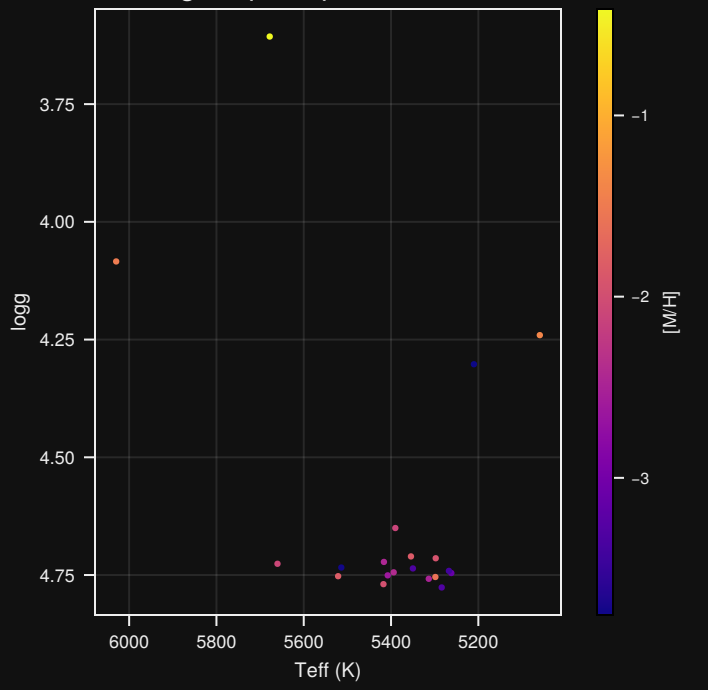
G0 & GSP-Phot coverage



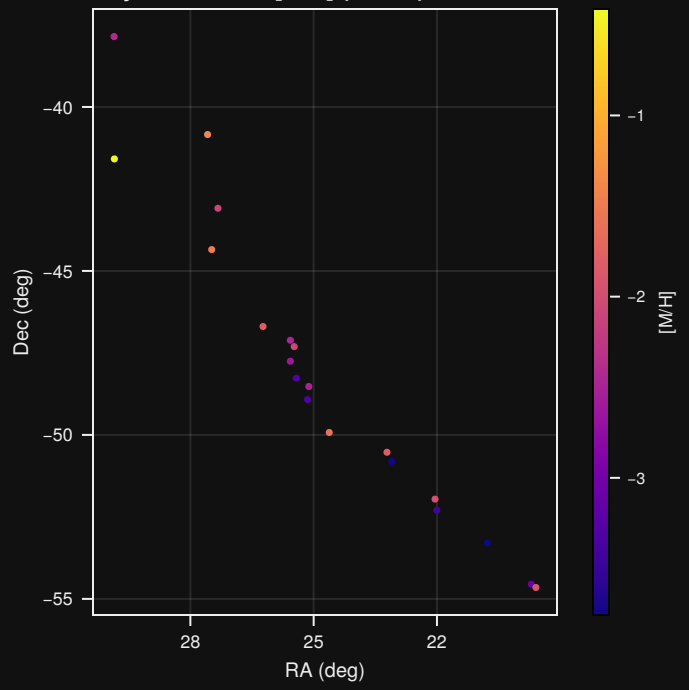
Star types from logg (n = 20)



Kiel diagram (n = 20)



Sky · GSP-Phot [M/H] (n = 20)

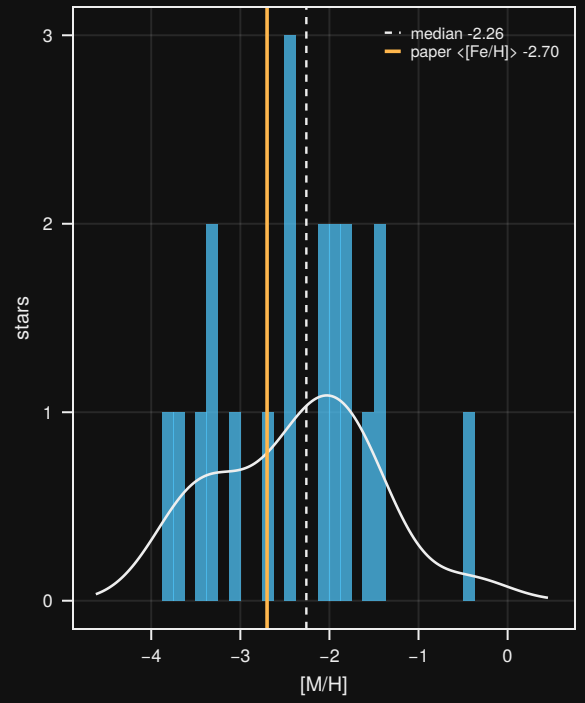


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

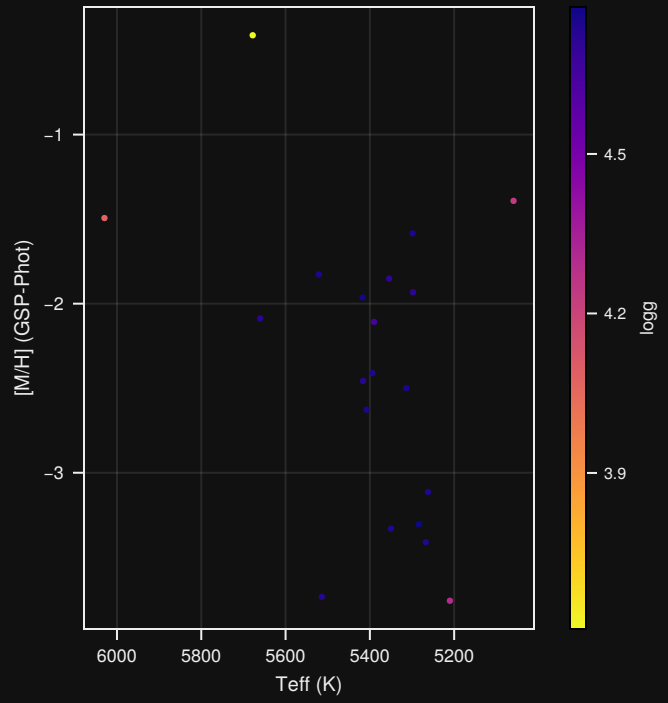
Phoenix

Metallicity · age · astrometric quality

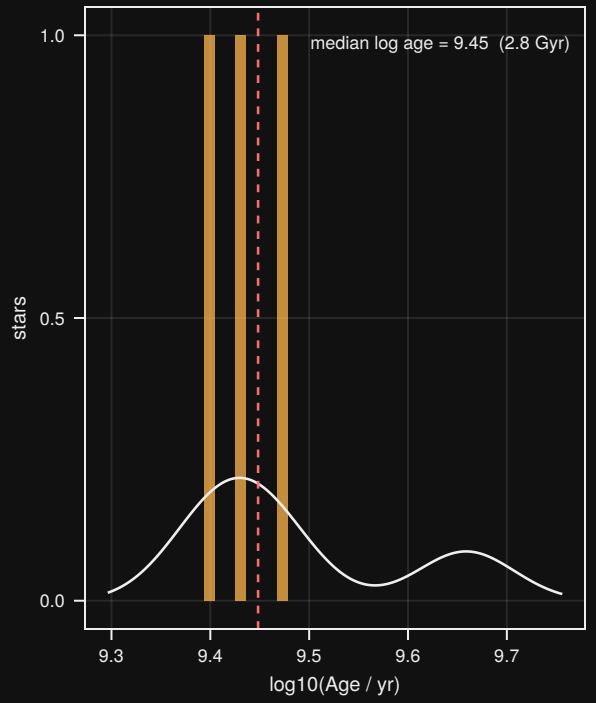
[M/H] GSP-Phot (n = 20)



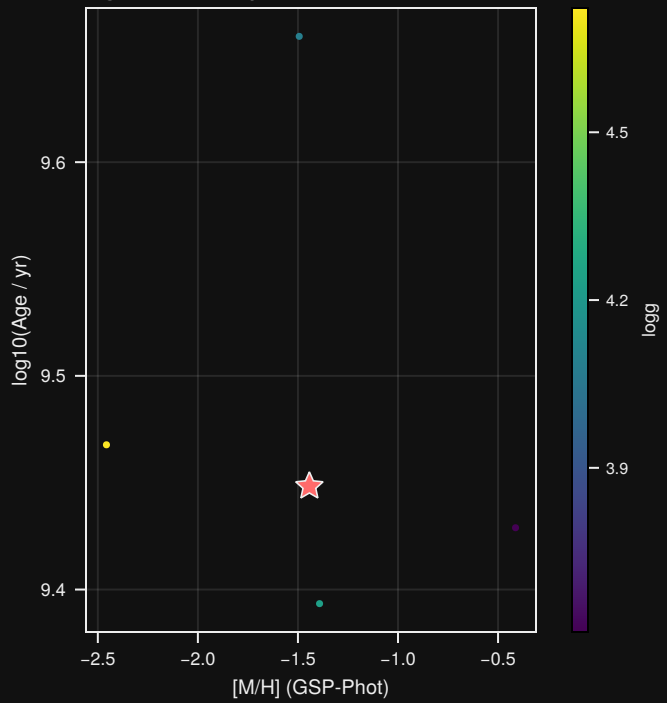
Teff vs [M/H] (n = 20)



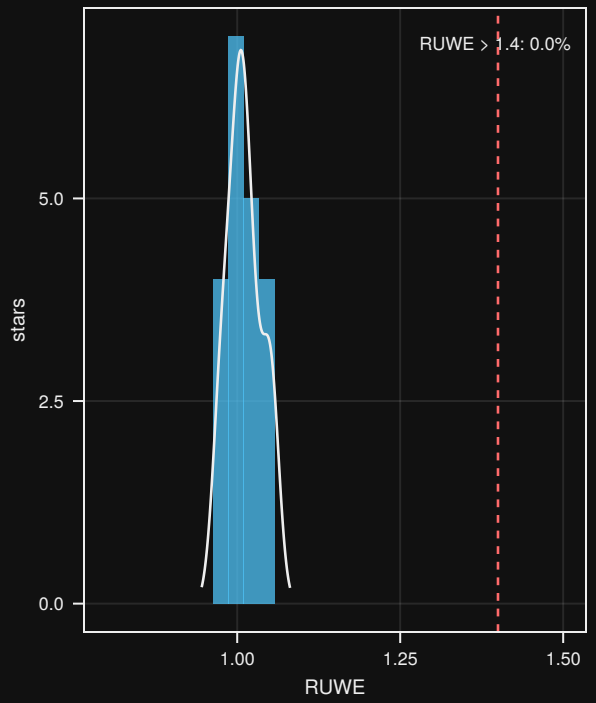
FLAME age (n = 4)



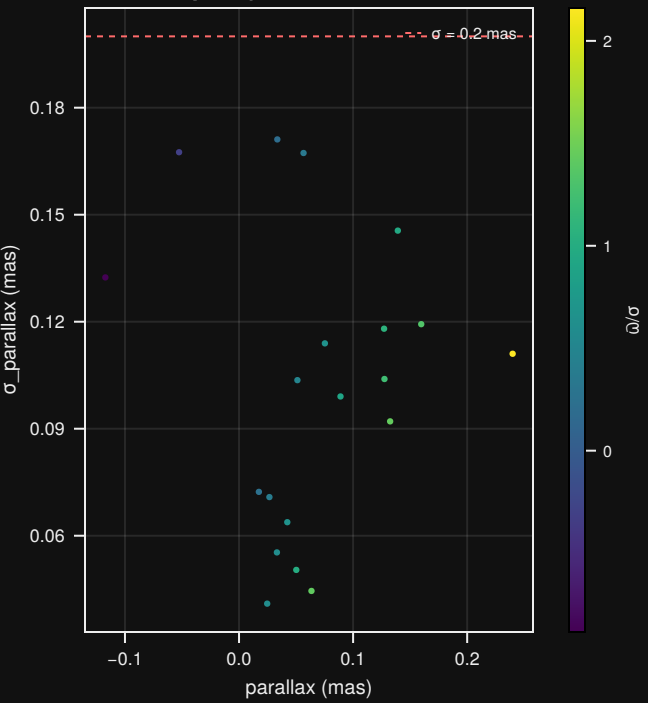
Age–metallicity (n = 4)



RUWE (n = 20)

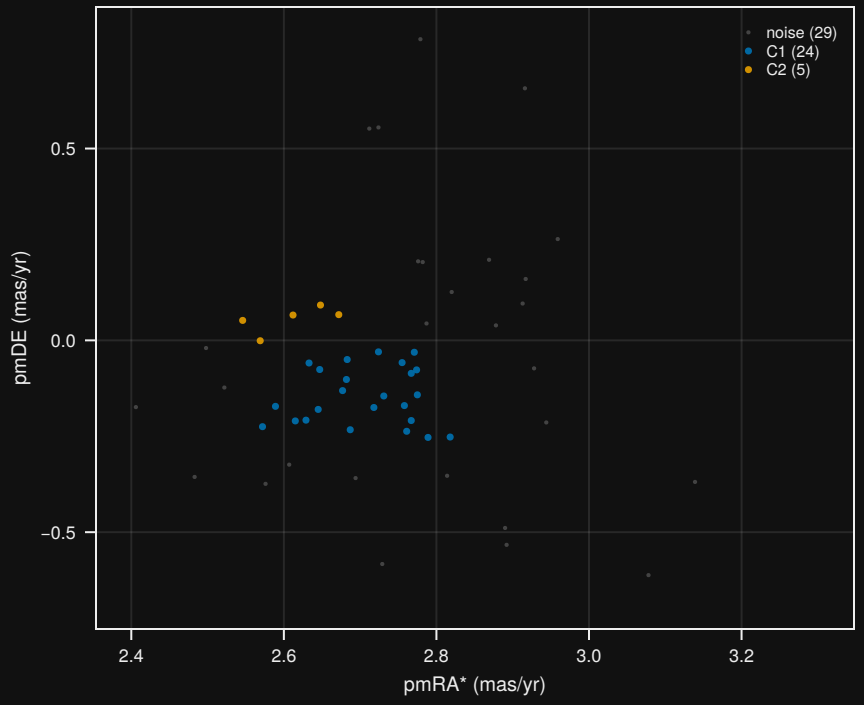


Parallax quality (n = 20)

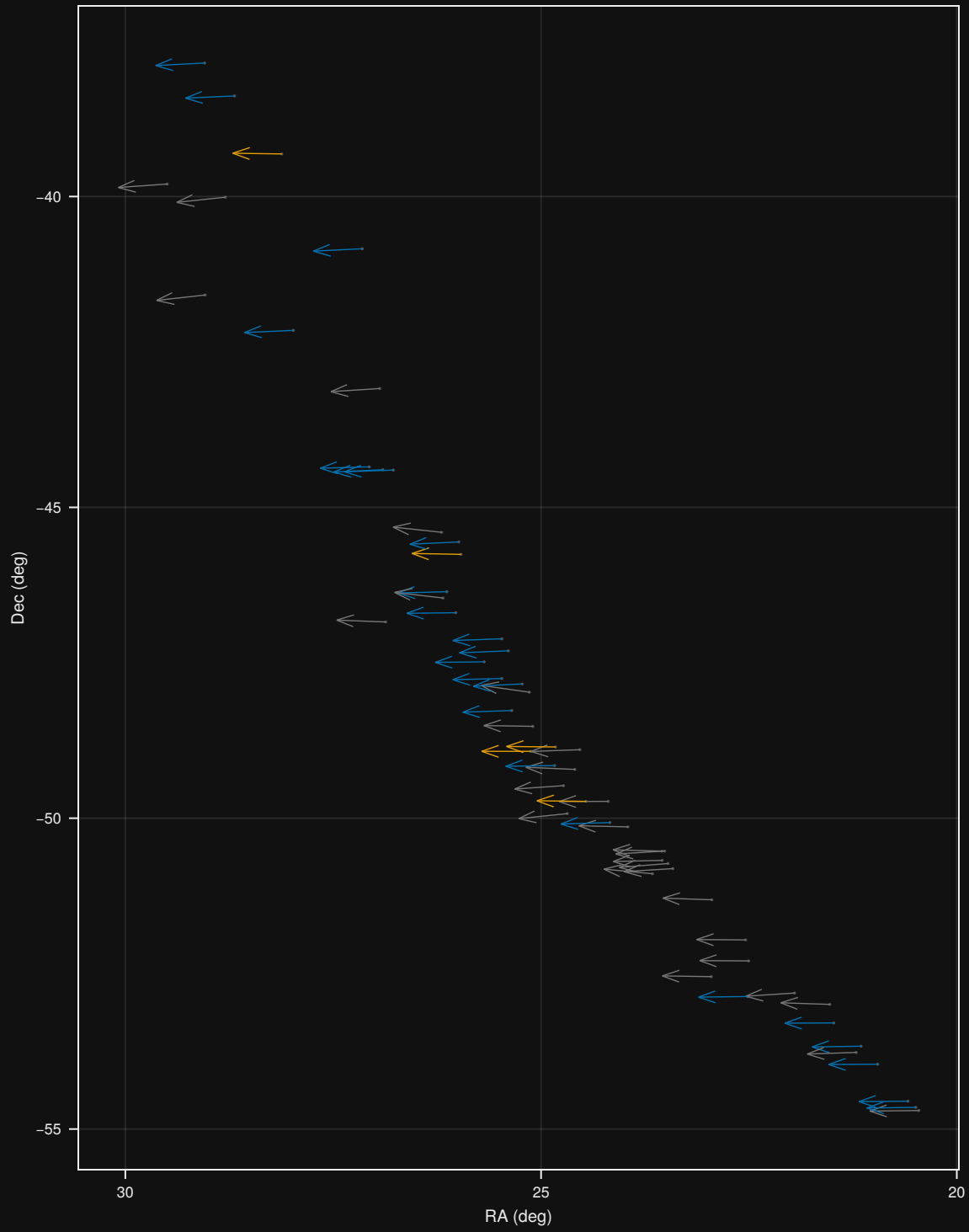


Phoenix

PM clustering · HDBSCAN · 2 cluster(s) · noise 50%



Proper-motion directions (arrows) coloured by cluster



Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	29	2.81	-0.02	0.15	-2.04	12
C1	24	2.72	-0.16	0.07	-2.41	13
C2	5	2.61	0.07	0.06	-3.33	1

Phoenix

Per-cluster proper-motion vector maps

